

R1

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1 New Code / Devs

1.1 Python Terminal Script for Filtering

Because we're likely to need to filter quite a few systems with very similar methods I decided to make an actual script instead of just using a notebook repetably. The largest upside is that this can easily be run on a compute node, which makes filtering larger sims easier.

Takes a user input in the CLI (see below), filters for all BH_Sol systems, and returns the population file with said system as well as a .csv of just the BH_Sol rows. Furthermore, while it is running it provides feedback via the terminal stating how many BH_MS systems it found, how many are the expected S1:S2 configuration. If any are S2:S1, it prints a warning as well as the total count.

Need to further refine it, but there's also a SBATCH script.

Usage Example

```
python BH-Sol_Filter.py /Data/1e+00_Zsun_population.h5 -o ./testResults
```

Arguments

Argument	Use	Ex value
-o	output location	.results
-outputName	name for the outputed files ¹	BH_Sol_Systems
-overwrite	Whether to overwrite existing results, T/F bool. Defaults to F	True
--maxMass	Max mass used when filtering, defaults to $3M_{\odot}$	3.5
--maxPeriod	Max period used when filtering, defaults to 3.5 days	4

Can be found at https://github.com/PiersonLip/stellarBHs/blob/main/FilteringScripts/BH-Sol_Filter.py

¹If a name is not specified it will use the original filename, appending a specifier to the front

2 New Research

2.1 Pop 1

When filtering via notebook and script, both returned 6 systems that had $p_{orb} < 3.5, M_{\odot,2} < 3, t < 10e9$

These systems all evolved via the same channel, that being ZAMS_oCE1_CC1_oRL02_CC2. Via inspection of the history df and the oneline, it is evident that the systems evolve detached, then post-MS **unstable** RLO happens via $S1 \rightarrow S2$. This leads to common envelope, which *greatly* decreases the orbital period. This envelope is then ejected, leaving a depleted stripped He core. The core then collapses in a couple thousand years, leaving the system in a BH-MS state with low p_{orb} /seperation.

These systems then evolve in a semi-peculiar way, where it is highly likely that there will be RLO from S2 onto S1, and because $S1 > S2$, this is stable, and leads to *very* prolonged stable RLO (seemingly on the scale of $> 10Gyr$), which leaves S2 at low masses ($< .08M_{\odot}$).

Interestingly, the initial period for all the systems is very close to $4000d$, with mean of 4022 and a std of only 151. However, a larger sample size is needed to confirm this.

Additionally, the initial masses of S1 and S2 appear to be quite constrained as well.

	orbital_period.i		S1_mass.i		S2_mass.i
count	6.000000	count	6.000000	count	6.000000
mean	4022.716197	mean	17.515761	mean	2.296788
std	151.522384	std	0.520269	std	0.320702
min	3873.118288	min	16.815111	min	1.797809
25%	3908.300983	25%	17.190036	25%	2.120753
50%	3978.305326	50%	17.510176	50%	2.353828
75%	4124.397101	75%	17.818155	75%	2.547210
max	4248.507680	max	18.254965	max	2.620529

2.2 Pop 2

Ran a second grid including a eccentricity distro, as well as slightly constraining the dataset further, hoping to confirm some of the init conditions

	orbital_period.i		S1_mass.i		S2_mass.i
count	12.000000	count	12.000000	count	12.000000
mean	4136.066721	mean	2.240382	mean	17.357478
std	177.676729	std	0.412075	std	1.073496
min	3844.570082	min	1.551266	min	15.506937
25%	4024.214908	25%	1.903135	25%	16.714083
50%	4119.397535	50%	2.300024	50%	16.998229
75%	4273.345845	75%	2.579111	75%	18.162633
max	4370.857404	max	2.729828	max	19.435080

2.3 Full P_{orb}

When filtering pop 2 with a max $p_{orb} = 300$ days we find 46 BH-Sol systems, all of these (except one!) evolve via the same CE channel. There is one system which does *not* do RLO before CC1, and it also only reaches a solar-like mass after RLO2, so I'm excluding it for now. Probably important to keep in mind though.

When plotting S1 vs S2 initial and formation masses there appears to be a correlation. In fig 2.3 this correlation appears to be quite strong, however, this very well may be because of the initial distribution.

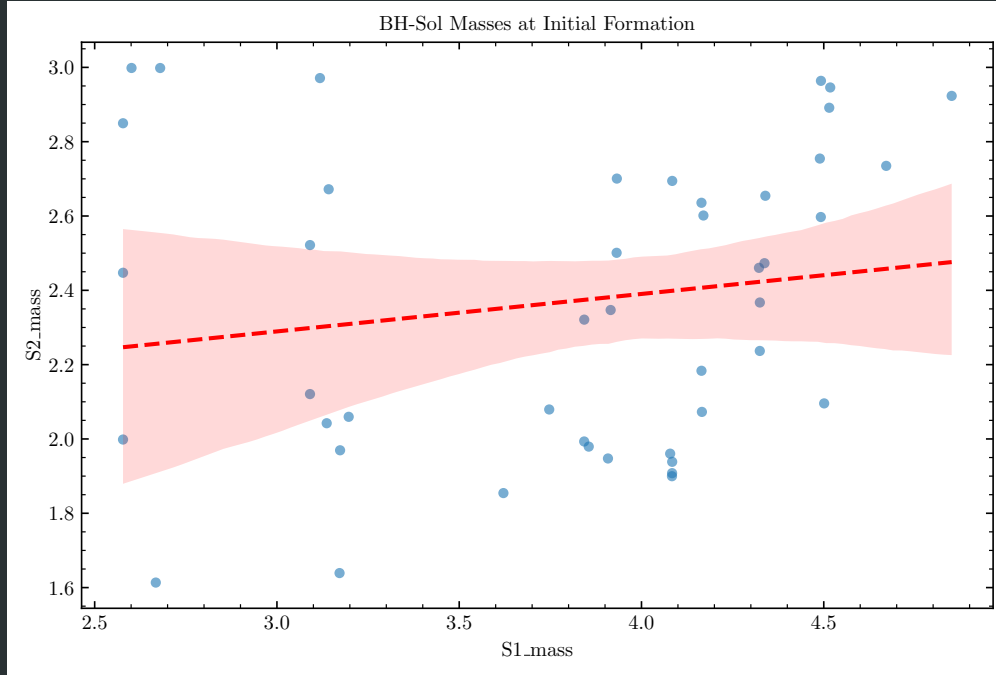


Figure 1: Masses at initial BH-Sol state. i.e. BH formation/S1 collapse

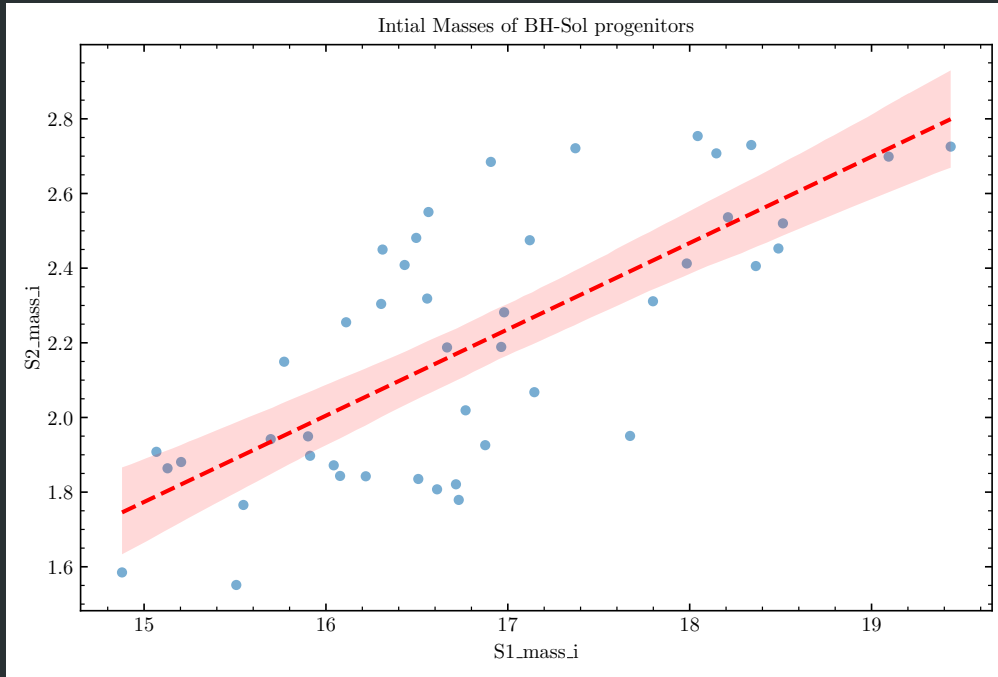


Figure 2: Initial ZAMS masses. May just be a byproduct of Ps and Qs, further sims with diff initial distribution needed.

3 To-Do

- Fid out if it makes sense for the onset of un-stable rlo to happen at $P_{orb} \approx 4000\text{days}$

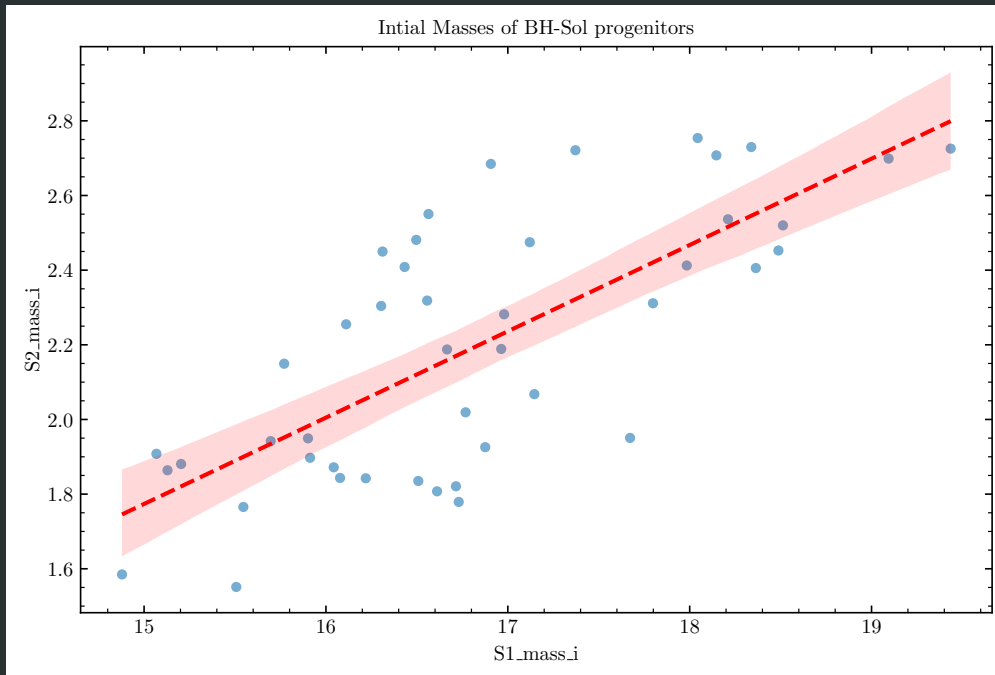


Figure 3: Initial ZAMS masses. May just be a byproduct of Ps and Qs, further sims with diff initial distribution needed.

4 Questions

- Tried running a grid, seems to have just completely failed from trying to refence itself before it was fully initalized. Entire log was filled with

```
Traceback (most recent call last):
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/popsyn/binarypopulation.py", 1
  binary.evolve()
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", 1
  self.run_step()
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/binarystar.py", 1
  next_step(self)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_merged.py
  super().__call__(binary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_isolated.
  super().__call__(binary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.
  self.update_after_evo(t, binary, primary, secondary)
File "/gpfs/projects/b1095/bku2126/UCXBInvestigation/POSYDON/posydon/binary_evol/DT/step_detached.
  current = history[-1]
          ~~~~~
IndexError: index -1 is out of bounds for axis 0 with size 0
```

and from what I can tell, this happened to nearly all the simmed bianaries

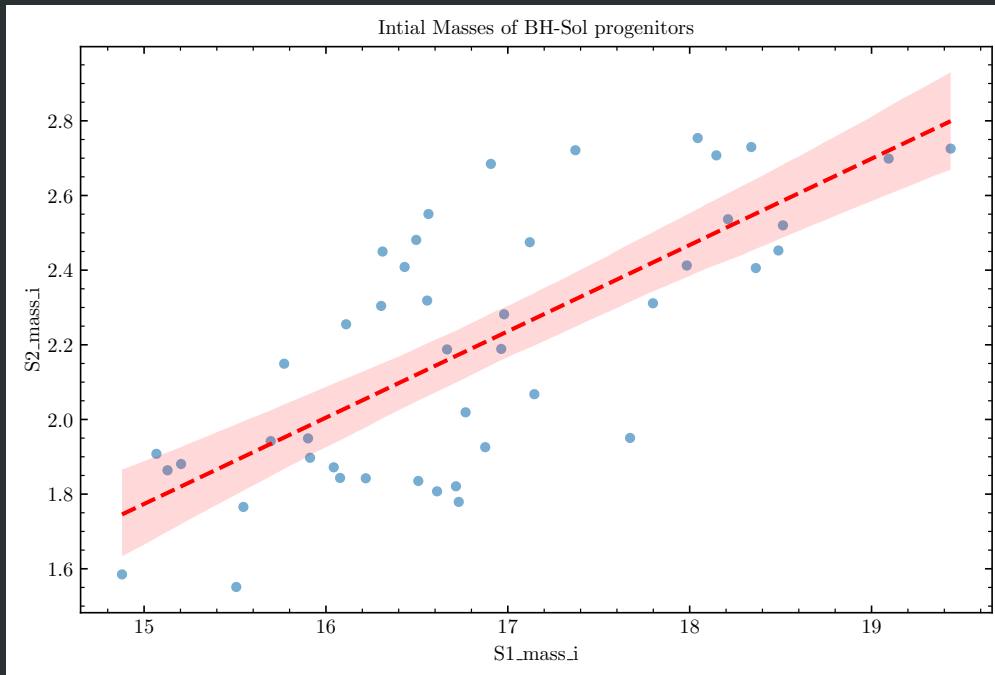


Figure 4: Initial ZAMS masses. May just be a byproduct of Ps and Qs, further sims with diff initial distribution needed.

- Should we include an initial eccentricity distribution? I imagine it would impact formation with unstable RLO and CE.
- Does it make sense to still be filtering for $p_{orb} < 3\text{days}$? The simulations seem to be suggesting that *all* BH-Sol systems evolve through CE, and that the main thing effecting p_{orb} post CE is the kick strength of the BH. So, maybe we should be initially filtering for all BH-Sol, and if we want the formation rate of specifically low p_{orb} ones, just take the subpop.
- Cool if i present an assortment of general POSYDON things ive worked on at astrofest?